



## LESSON 7.7

# SOLVING MULTI-STEP PROBLEMS INVOLVING DISCOUNTS, TAXES, TIPS, OR FEES

## LESSON INTRODUCTION

**MA.7.AR.3.1** Apply previous understanding of percentages and ratios to solve multistep, real-world percent problems.

## LEARNING TARGETS

- I can solve multistep problems that involve discounts, markups, taxes, tips, or fees.

## BENCHMARK COHERENCE

### BUILDING ON

**MA.6.AR.3.4** Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

### WORKING TOWARD

N/A

## ESSENTIAL QUESTIONS

- How do you solve multistep problems that involve discounts, markups, taxes, tips, or fees?
- How can you determine the amount of sales tax or percent gratuity given a bill subtotal and final total?

## LESSON NARRATIVE

In this lesson, students solve multistep real-world problems involving discounts, markups, taxes, tips, or fees. This lesson builds on students' prior work in Lessons 5-6 solving one-step discount, markup, tax, tip, and fee problems.

## COMPONENT SUMMARY

### 7.7.1 WARM-UP (~5 MIN.)

Consider percentages on a double number-line model

### 7.7.2 EXPLORATION (10-15 MIN.)

Explore finding sales tax rates and tipping in real-world contexts

### 7.7.3 COLLABORATION (10-12 MIN.)

Collaboratively solve problems involving taxes, tips, and totals

### 7.7.4 YOUR TURN! (10-13 MIN.)

Practice solving problems involving taxes, tips, fees, discounts, and markups

### 7.7.5 WRAP-UP (~5 MIN.)

Demonstrate understanding of solving multistep sales tax, discount, markup, and fee problems

## RESOURCES

### GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Gratuity

### MATERIALS

- (Optional) Chart paper & markers

### ADDITIONAL PREPARATION

- Consider what calculations/work you want students to show in their workbook for problems where they use a calculator.

## TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may struggle converting between different forms of rational numbers (i.e., fraction to decimal, percent to decimal, decimal to percent).
- Students may not understand the difference between the subtotal and the total and which are both used when determining tax or tip amounts in various contexts.

### THINGS TO CONSIDER

- During the 7.7.2 Exploration, consider modeling a portion of problem if needed, providing specific instructions on strategies that can be used to find a sales tax percentage. Listen as students collaborate on question 1a to determine if this is necessary.
- During the 7.7.4 Your Turn!, facilitate a discussion with students explaining that tips can be based on the subtotal (without tax) or based off the total (with taxes).
- Allowing students to use calculators during the lesson will help students focus on discovery and process of calculating discounts, markups, taxes, tips, and percentages. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers.
- There are multiple strategies for solving the problems in this lesson. Encourage students to use strategies that make sense for them as they build fluency in solving problems.

### LITERACY AND LANGUAGE ROUTINES

#### Think Pair Share

In 7.7.3 Collaboration, consider asking students to initially complete the problems independently. Then have students discuss their work with a partner. These discussions will help students gain confidence and understanding.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

#### MA.K12.MTR.6.1 Reasonableness

In 7.7.4 Your Turn!, students solve multistep problems using a variety of strategies. Prompt students to informally check their answers by considering if the solution makes sense given the context. Then have students also formally check their calculations for accuracy.

## DIFFERENTIATE INSTRUCTION

Consider building upon any anchor charts created in earlier lessons to add additional content learned.

Consider modifying the 7.7.3 Collaboration and 7.7.4 Your Turn! into stations that students can work through with a partner or small group. Determine how to set up the rotation and which stations may require more support from the teacher as students work through them.

## 7.7.1 WARM-UP: PERCENTAGES ON A LINEAR MODEL

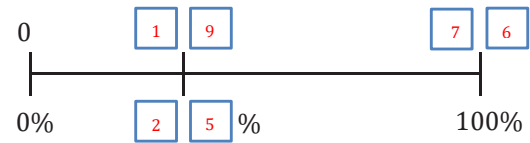
### TEACHER MOVES

Students have worked with double number-line models before. If time allows, consider having students discuss question 2b and/or share strategies/solutions for question 1.



### 7.7.1 Warm-Up: Percentages on a Linear Model

1. Fill in the blue boxes using the digits 1 – 9, one time each at most, to create an accurate number line.



2. There are multiple possible solutions.
- Find at least one other solution.  
*17, 68, 25%*
  - Explain whether using 47% for the percentage makes sense.  
*No, because the line would need to be closer to the middle since 47 is close to 50 which is halfway.*



### 7.7.2 Exploration: Finding Sales Tax Rates and Total Amounts

1. Cities often have different sales tax rates based on local government decisions.

- a. Discuss with your partner how you might find the percentage of sales tax if you are given the price of an item and the amount of sales tax paid for the item. Summarize your discussion.

*If you divide the sales tax by the price, you can find the rate. You could also set up a percent proportion to find the rate.*

- b. With your partner, complete the two tables to determine the sales tax rates for City A and City B.

| City A       |            |                |                    |
|--------------|------------|----------------|--------------------|
| Item         | Price (\$) | Sales Tax (\$) | Sales Tax Rate (%) |
| Paper Towels | 8.99       | 0.63           | 7%                 |
| Laundry Soap | 11.83      | 0.83           | 7%                 |

| City B       |            |                |                    |
|--------------|------------|----------------|--------------------|
| Item         | Price (\$) | Sales Tax (\$) | Sales Tax Rate (%) |
| Paper Towels | 8.99       | 0.54           | 6%                 |
| Laundry Soap | 11.83      | 0.71           | 6%                 |

- c. Determine the total cost, with tax, of paper towels and laundry soap in City A.  
\$22.28
  - d. The pre-tax cost of 5 beach towels in City A is \$54.95. Determine the total cost, with tax, of 5 beach towels in City A.  
\$58.80

## 7.7.2 EXPLORATION: FINDING SALES TAX RATES AND TOTAL AMOUNTS

### DIFFERENTIATE INSTRUCTION

If needed, consider doing a Think Aloud as you solve question 1b (only City A) after giving students an opportunity to discuss question 1a. This will provide a model for students as they try the other problems.

### ENRICHMENT

- Is the sales tax rate for both items in one city the same or different?
- Why would this make sense in context?

## 7.7.2 EXPLORATION: FINDING SALES TAX RATES AND TOTAL AMOUNTS (CONTINUED)

### TEACHER MOVES

Prompt students to calculate the tip and then compare their answers to Jada's and Corrin's answers. Remind students to convert the tip percentage to a decimal.

- e. If the subtotal for the purchase of a new bathing suit in City B is \$47.97, what is the total cost of the bathing suit with tax?  
\$50.85

- f. Find the total cost, with tax, on paper towels, laundry soap, and a new bathing suit in City B.  
\$72.92

2. Jada and Corrin had lunch at a local restaurant. The final bill for their meal is shown.

They decide to tip their server 18% of the subtotal. Jada says the tip should be \$7.56, but Corrin disagrees. Corrin calculated the tip to be \$8.28.

- a. Explain who calculated the correct tip.

*Jada correctly calculated the tip because \$42.00 times 0.18 is \$7.56. Corrin calculated the tip from the total which includes taxes, instead of from the subtotal.*

- b. What is the total amount, including the tip, that the friends paid at the restaurant?  
\$53.55

Date: Sep. 12th  
Time: 6:55 PM  
Server: # 27

|              |       |
|--------------|-------|
| Bread Stix   | 9.50  |
| Chicken Parm | 15.50 |
| Chef Salad   | 12.00 |
| Lemon Soda   | 2.00  |
| Tea          | 3.00  |

|           |       |
|-----------|-------|
| Subtotal  | 42.00 |
| Sales Tax | 3.99  |
| Total     | 45.99 |



### 7.7.3 Collaboration: Taxes, Tips, and Totals

Work through each situation with your partner. Be sure to show your work.

1. Xavier visits an Italian restaurant for dinner.

- a. If Xavier tips 20% on the subtotal \$24.95, how much tip will he pay?  
**\$4.99**

Date: Sep 19th  
Time: 6:04 PM  
Server: Flo

Bread Stix 9.50  
Ravioli Bites 10.50  
Cheesecake 4.95

Subtotal 24.95  
Sales Tax \_\_\_\_\_  
Tip \_\_\_\_\_  
Total \_\_\_\_\_

- b. The city has a sales tax rate of 9.5%. How much in tax will the customer pay on the subtotal of \$24.95?  
**\$2.37**
- c. What is the total Xavier spent on his meal, including tip and tax?  
**\$32.31**

2. A sporting goods store purchased baseball hats from a supplier for \$24 each and marked them up to \$35.
- a. Calculate the percentage of the markup.  
**46%**
- b. A customer orders 4 baseball hats from the store online. What is the subtotal of the 4 hats?  
**\$140.00**
- c. If the sales tax is 6.25% of the subtotal and there is a \$7.00 shipping fee applied after tax, what is the total amount the customer paid?  
**\$155.75**
3. Leilani found a 30% off discount code for an NXT True Hybrid Surfboard. The original cost of the surfboard is \$700. If the company does not charge tax but instead charges a \$150 shipping fee, how much does the surfboard cost in total?  
**\$640.00**

## 7.7.3 COLLABORATION: TAXES, TIPS, AND TOTALS

### TEACHER MOVES

#### Think-Pair-Share

Ask students to initially complete the questions independently. Then have students discuss their work with a partner. These discussions will help students gain confidence and understanding.

### ENRICHMENT

Consider asking students the following:

- Does the sales tax amount depend on the tip amount? Or are those two separate additional costs?
- Are these amounts related to the subtotal or the total cost?

### DIFFERENTIATE INSTRUCTION

Consider having students write their work on individual dry-erase boards to make sharing their work during the debrief easier.

## 7.7.4 YOUR TURN!

### DIFFERENTIATE INSTRUCTION

Consider combining the 7.7.3 Collaboration and Your Turn! into stations and having students rotate through the problems.



### 7.7.4 Your Turn!

- A car dealership pays a wholesale price of \$12,000 to purchase a vehicle. The car dealership wants to make a 32% profit.
 
  - How much money is added to the wholesale price to make a 32% profit?  
**\$3,840**
  - After the markup, what is the total retail price of the vehicle?  
**\$15,840**
  - The sales tax in the city it is sold is 7.5%. After tax, what will the customer pay for the car?  
**\$17,028**
- The table shows the pre-tax cost of items at a convenience store where the sales tax rate is 4.6%.

| Item                 | Cost (\$) |
|----------------------|-----------|
| Milk                 | 3.75      |
| Bag of Ice           | 2.50      |
| Aspirin – 24 tablets | 8.25      |
| Snack Mix            | 3.75      |
| Candy Bar            | 1.25      |
| Batteries            | 6.75      |

- Cal stopped at the convenience store and picked up a bag of ice and batteries. What was his total cost, with tax?  
**\$9.68**

### ENRICHMENT

Consider asking students the following:

- What is gratuity? What does it mean that gratuity is pretax?
- How can you find the sales tax percentage from the sales tax and subtotal?
- Can you find the sales tax and gratuity separately?

- Ms. Isales has \$10.00 in her wallet. Does she have enough to buy aspirin and a candy bar? Explain your reasoning using mathematics.  
**Yes, because when you add the items and find the tax it adds up to \$9.94 which is less than \$10.**
- A local restaurant automatically adds a 21% gratuity (tip), pre-tax, to each bill for parties over 10. The table shows the bills due for several parties over 10. Complete the table.

| Party          | Subtotal (\$) | Sales Tax (\$) | Gratuity (\$)  | Total Amount of Bill |
|----------------|---------------|----------------|----------------|----------------------|
| Flores Party   | 93.75         | 5.86           | <b>\$19.69</b> | <b>\$119.30</b>      |
| Atkinson Party | 176.99        | <b>\$11.06</b> | 37.17          | <b>\$225.22</b>      |
| Mikelson Party | 135.50        | <b>\$8.47</b>  | <b>\$28.46</b> | 172.43               |

**7.7.5 Wrap-Up: Ticket Out the Door**

1. Diesel has been waiting patiently for the Evolve Carbon All Terrain Skateboard to go on sale. For his birthday, he received a 25% off discount. The original price of the skateboard is \$1,999.99 and the company has a 5% sales tax, along with a flat fee of \$35 for shipping.

What is the total amount Diesel needs to purchase his dream skateboard?

**\$1,609.99**

**7.7.5 WRAP-UP: TICKET OUT THE DOOR****QUESTIONING**

Challenge students who finish early to write their own multistep problem. Then have students exchange problems with a partner and solve each other's problem.